

UCH-ADZHI, Turkmenistan

WMO: 387990

Lat: 38.083N

Lon: 62.800E

Elev: 607

StdP: 14.38

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	19.1	4.0	6.9	16.2	9.4	9.1	26.1	22.9	56.1	20.6	50.7	4.0	110	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.1	109.2	71.3	106.7	70.8	104.3	70.0	75.3	104.1	73.6	102.5	72.0	100.7	10.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	94.2	93.5	62.6	86.7	90.9	60.4	80.1	89.4	39.2	104.1	37.5	102.5	36.1	100.8	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	16.5	14.3	7.2	113.6	8.6	2.0	1.0	115.0	-4.0	116.2	-8.9	117.3	-15.1	118.8
			6.6	76.4	8.5	4.0	0.4	79.2	-4.6	81.6	-9.3	83.8	-15.5	86.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	39.2	42.9	53.9	65.7	77.4	86.3	90.1	86.6	75.1	61.9	48.6	40.5	(6)
(7)		DBStd	19.89	9.92	10.34	9.32	8.99	6.69	5.47	4.57	5.56	6.88	8.94	9.70	8.92	(7)
(8)		HDD50	1122	352	237	66	4	0	0	0	0	14	135	314		(8)
(9)		HDD65	3310	799	620	360	100	4	0	0	8	164	497	758		(9)
(10)		CDD50	6288	18	37	187	477	848	1090	1244	1135	753	384	94	21	(10)
(11)		CDD65	2999	0	0	16	122	387	640	779	670	310	69	6	0	(11)
(12)		CDH74	46753	1	7	231	1571	5435	10437	13075	10504	4335	1057	98	2	(12)
(13)		CDH80	29488	0	0	65	686	3046	6912	9006	6979	2365	401	28	0	(13)
(14)	Wind	WSAvg	6.5	6.8	7.2	7.1	6.6	6.4	7.0	7.6	6.9	5.5	5.2	5.4	6.3	(14)
(15)	Precipitation	PrecAvg	4.6	0.7	0.8	1.1	0.6	0.3	0.0	0.0	0.0	0.2	0.6	0.5		(15)
(16)		PrecMax	7.4	2.2	2.8	3.6	1.5	1.5	0.3	0.1	0.0	0.4	1.0	1.3	1.6	(16)
(17)		PrecMin	1.7	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.4	0.5	0.7	0.7	0.4	0.4	0.1	0.0	0.1	0.2	0.4	0.4		(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.1	73.8	87.3	98.9	104.5	110.5	113.3	111.0	103.8	95.6	83.7	70.6	(19)
MCWB			51.5	54.7	60.7	67.2	70.2	71.6	72.0	73.1	69.3	65.4	60.0	51.6	(20)	
2%		DB	63.3	67.8	80.3	92.5	100.7	107.2	110.2	107.7	99.9	88.7	75.7	64.9	(21)	
		MCWB	48.5	51.6	58.8	65.5	68.4	70.8	70.8	71.5	67.8	63.0	56.5	49.7	(22)	
5%		DB	58.2	62.8	75.1	87.7	97.6	104.7	107.6	105.0	96.3	84.2	70.1	59.4	(23)	
		MCWB	46.7	49.8	56.5	63.1	67.0	69.8	70.9	70.5	66.5	61.3	53.7	47.4	(24)	
10%		DB	53.2	58.4	70.0	83.2	94.2	102.3	104.8	102.2	92.4	79.8	65.1	54.6	(25)	
		MCWB	44.5	47.9	54.6	61.1	65.7	69.1	70.3	69.4	64.9	58.9	51.7	45.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	55.8	62.7	70.1	73.7	77.1	78.7	76.2	72.7	67.8	61.7	53.4	(27)
MCD B			66.1	68.6	82.0	92.7	100.1	105.3	107.7	105.4	98.7	91.5	78.5	66.9	(28)	
2%		WB	50.3	53.4	60.0	67.0	71.0	74.2	75.7	74.3	70.2	64.4	58.0	50.7	(29)	
		MCD B	60.5	64.6	76.3	88.2	95.8	101.9	104.4	104.3	96.3	85.4	72.0	62.4	(30)	
5%		WB	47.7	51.0	58.0	64.5	68.8	72.4	73.8	72.5	67.9	61.9	55.0	48.3	(31)	
		MCD B	56.4	60.8	72.6	83.1	92.2	100.7	103.0	102.2	93.3	81.6	67.6	57.6	(32)	
10%		WB	44.9	48.5	55.6	62.4	66.8	70.5	72.0	70.7	65.4	59.6	52.6	46.0	(33)	
		MCD B	52.7	57.5	68.0	80.2	89.9	99.0	101.1	99.9	90.6	78.1	63.5	54.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.4	19.8	22.0	24.5	26.8	27.2	26.1	27.7	30.2	29.3	23.5	18.5	(35)
MCDBR			25.7	26.8	30.5	31.8	32.0	30.7	30.0	30.3	34.5	34.0	30.4	26.6	(36)	
MCWBR			13.5	13.2	12.5	12.8	12.8	12.9	12.1	14.1	17.1	17.5	15.6	14.4	(37)	
5% WB		MCDBR	22.9	23.5	27.1	28.0	28.1	27.4	27.2	28.2	31.6	31.3	27.2	24.5	(38)	
		MCWBR	12.9	12.4	11.9	11.9	13.1	14.1	14.2	15.1	17.9	17.0	14.4	13.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.389	0.433	0.506	0.535	0.534	0.542	0.521	0.476	0.450	0.474	0.440	0.392	(40)	
taud		2.089	1.979	1.832	1.804	1.827	1.802	1.853	1.949	1.980	1.901	1.971	2.072	(41)		
Ebn at Noon		239	243	237	240	243	241	245	253	251	226	216	226	(42)		
Edh at Noon		38	48	61	67	66	68	64	57	52	51	42	36	(43)		
(44)	All-Sky Solar Radiation	RadAvg	701	1007	1373	1815	2238	2528	2490	2299	1899	1334	859	643	(44)	
(45)		RadStd	62	106	101	140	87	75	53	56	48	78	77	70	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-2.84	-2.09	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	-0.97	-0.86	N/A	N/A	N/A	+162

Nomenclature: See separate page